

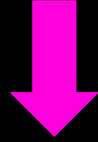
FREQUENCY RESPONSE OF BJT

Er. Sailesh Mishra

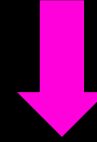
B.Tech (ECE | Silicon University)

Specialization in Microelectronics & VLSI

FREQUENCY RESPONSE

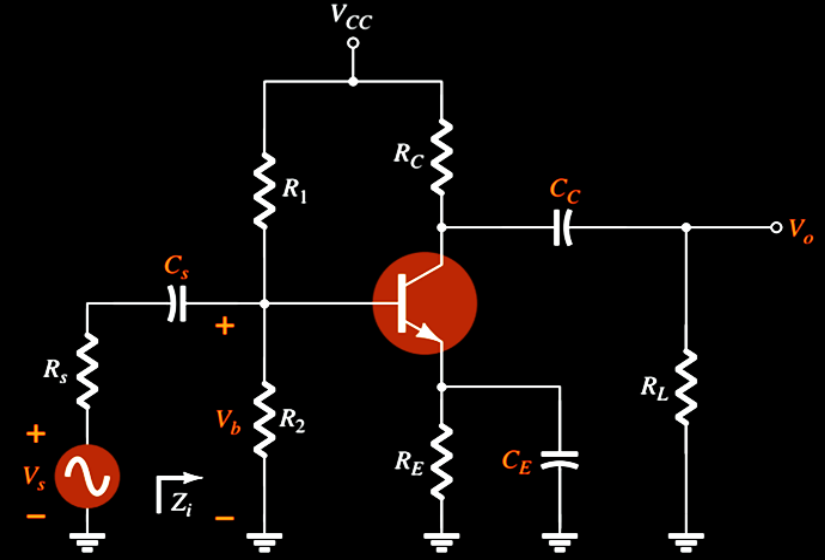


**LOW
FREQUENCY RESPONSE**

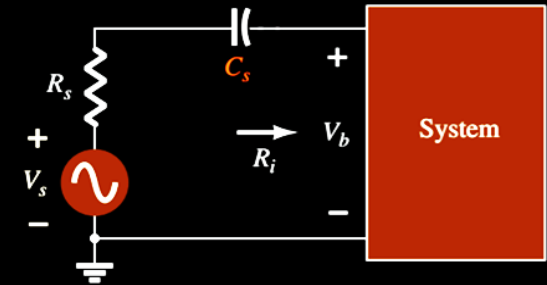
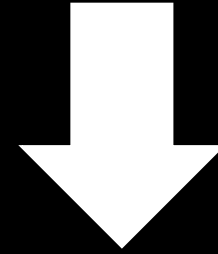


**HIGH
FREQUENCY RESPONSE**

LOW FREQUENCY RESPONSE

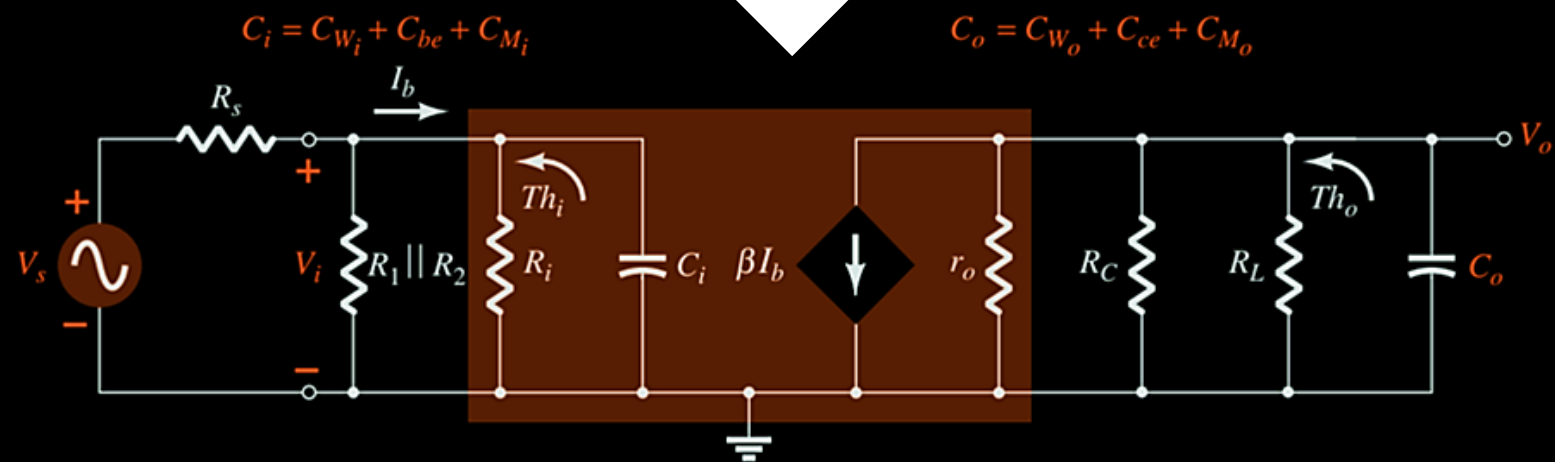
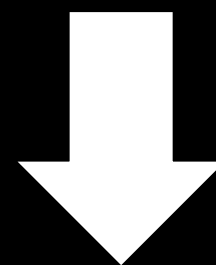
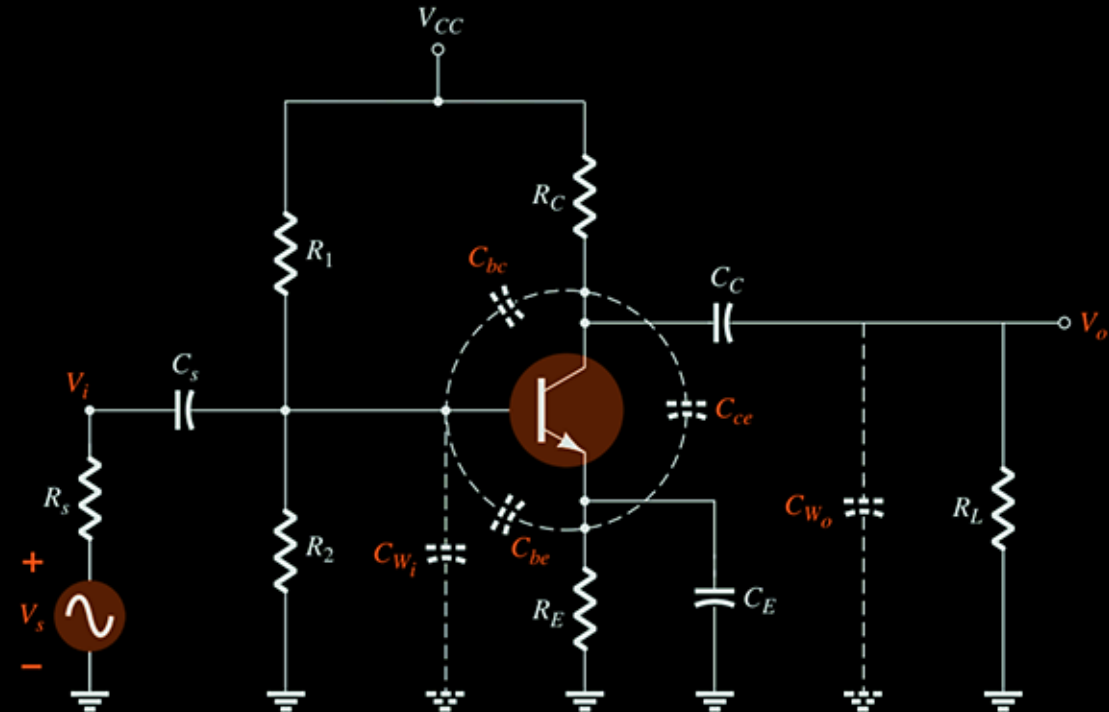


Determining the effect of R_s on the low-frequency response of a BJT amplifier.

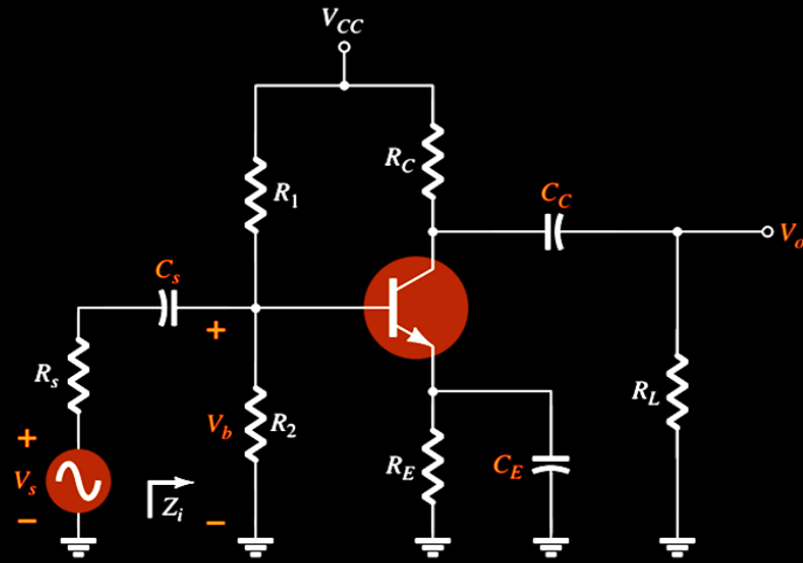


Determining the effect of C_s on the low-frequency response.

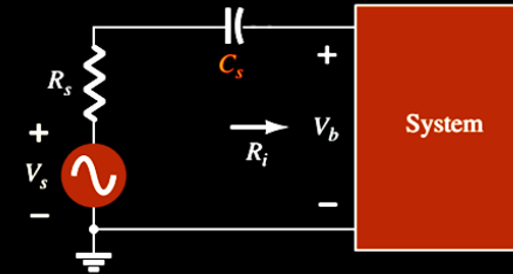
HIGH FREQUENCY RESPONSE



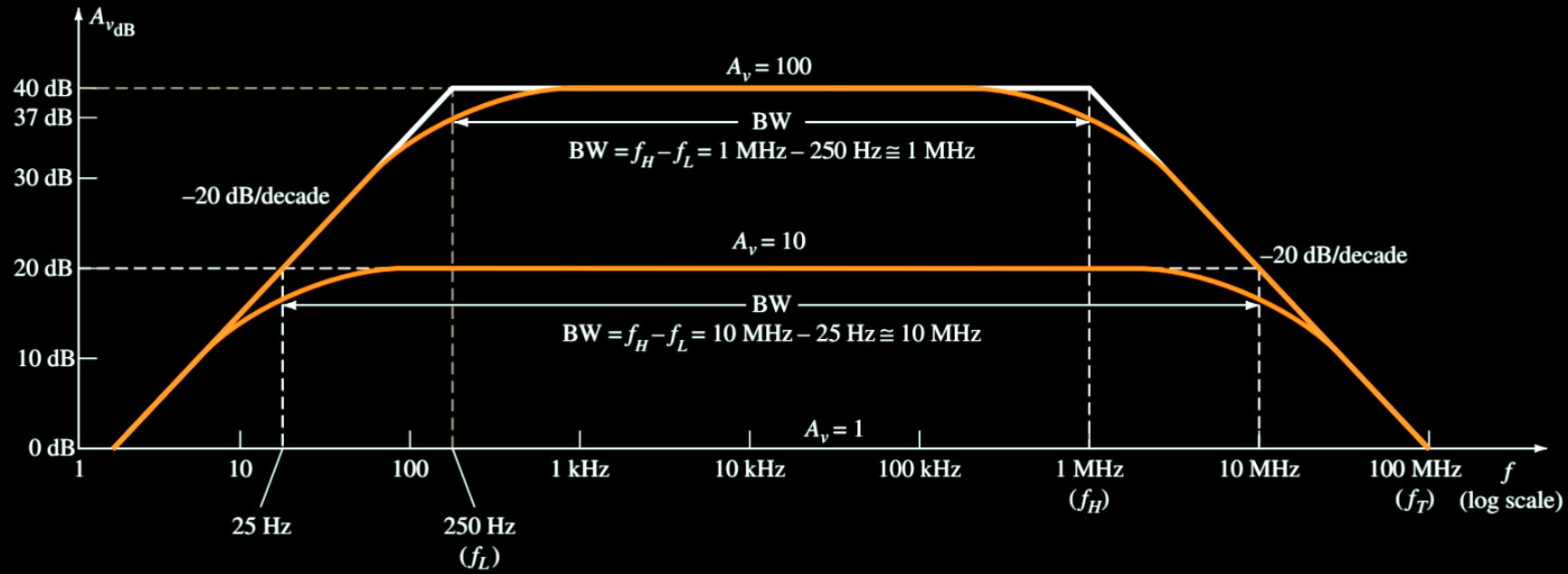
FREQUENCY RESPONSE OF BJT AMPLIFIER

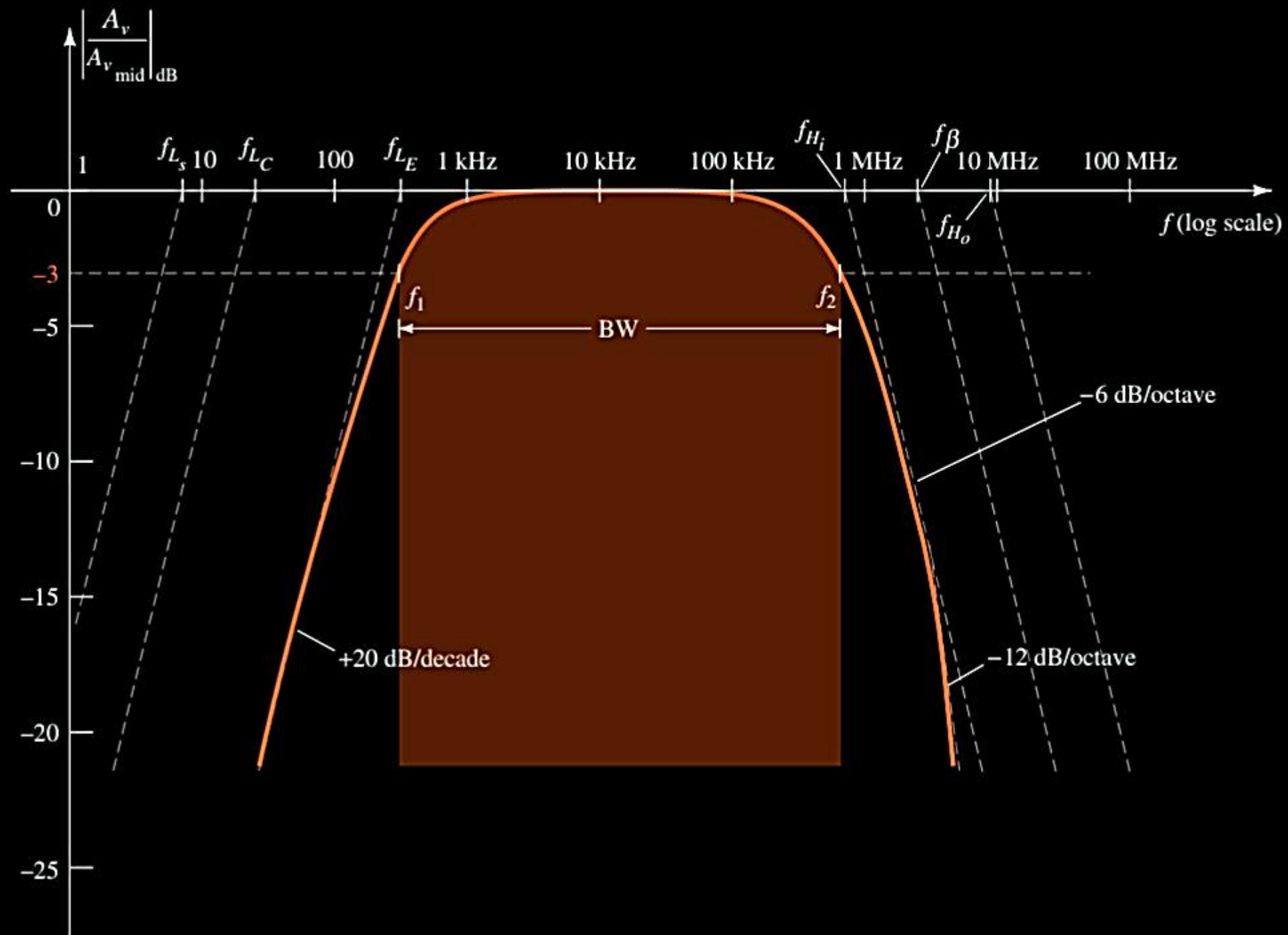


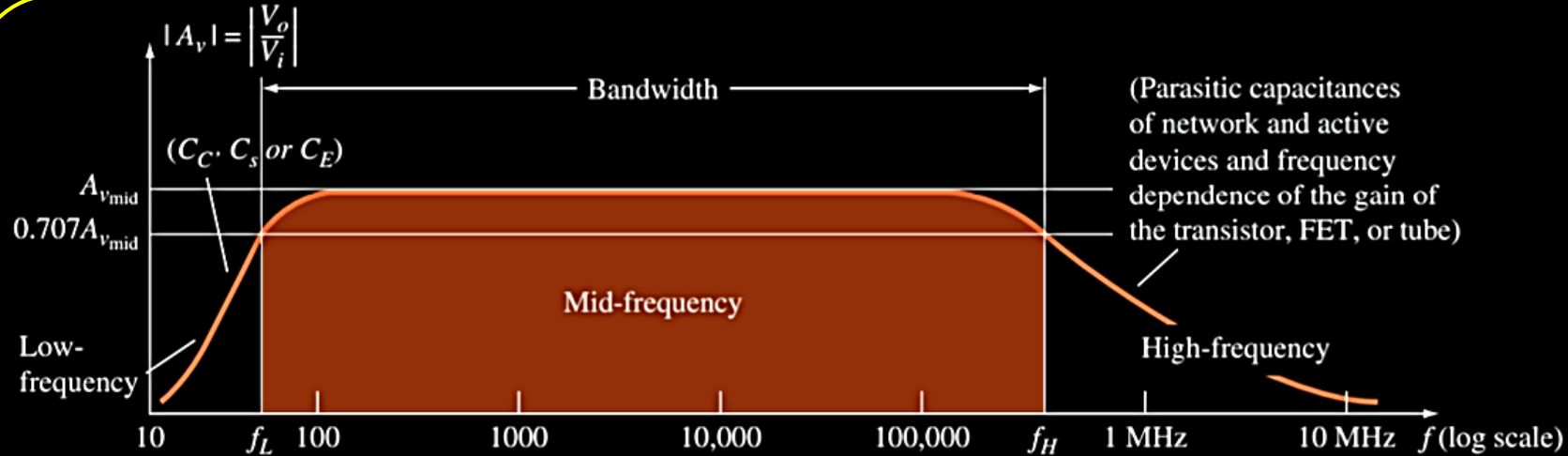
Determining the effect of R_s on the low-frequency response of a BJT amplifier.



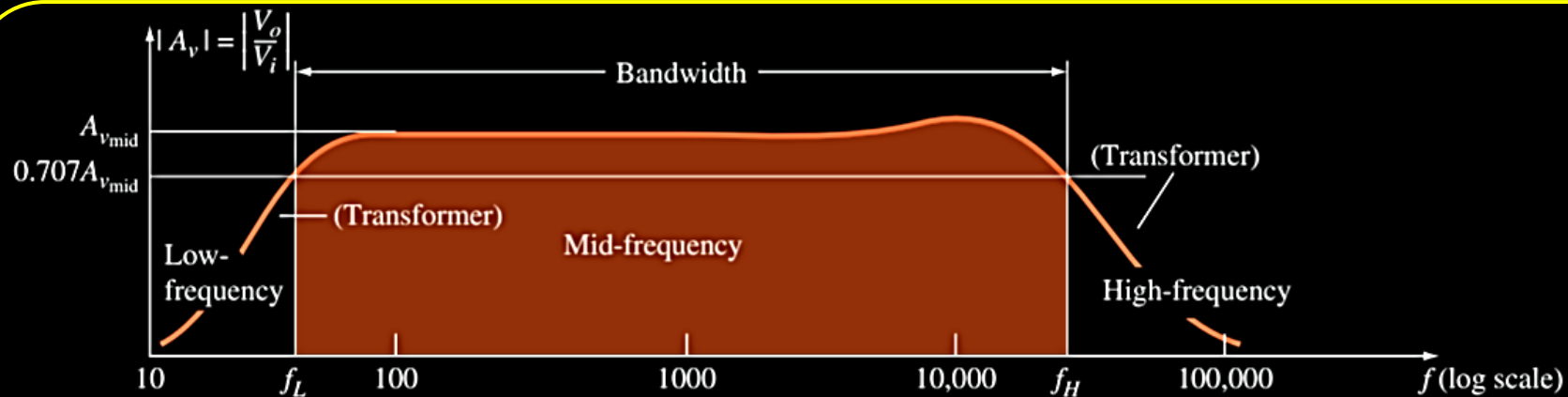
Determining the effect of C_s on the low-frequency response.



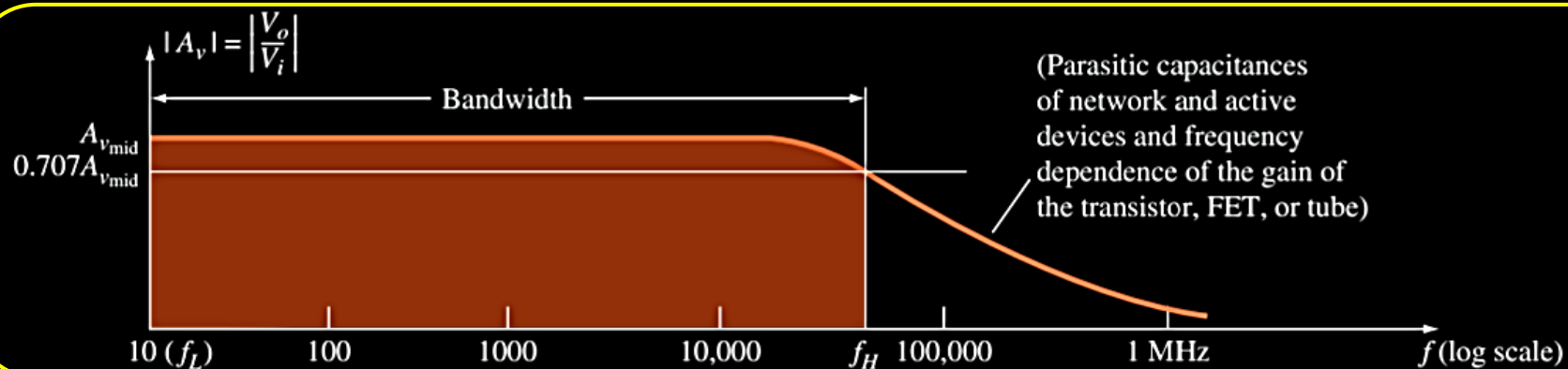




RC COUPLED AMPLIFIER

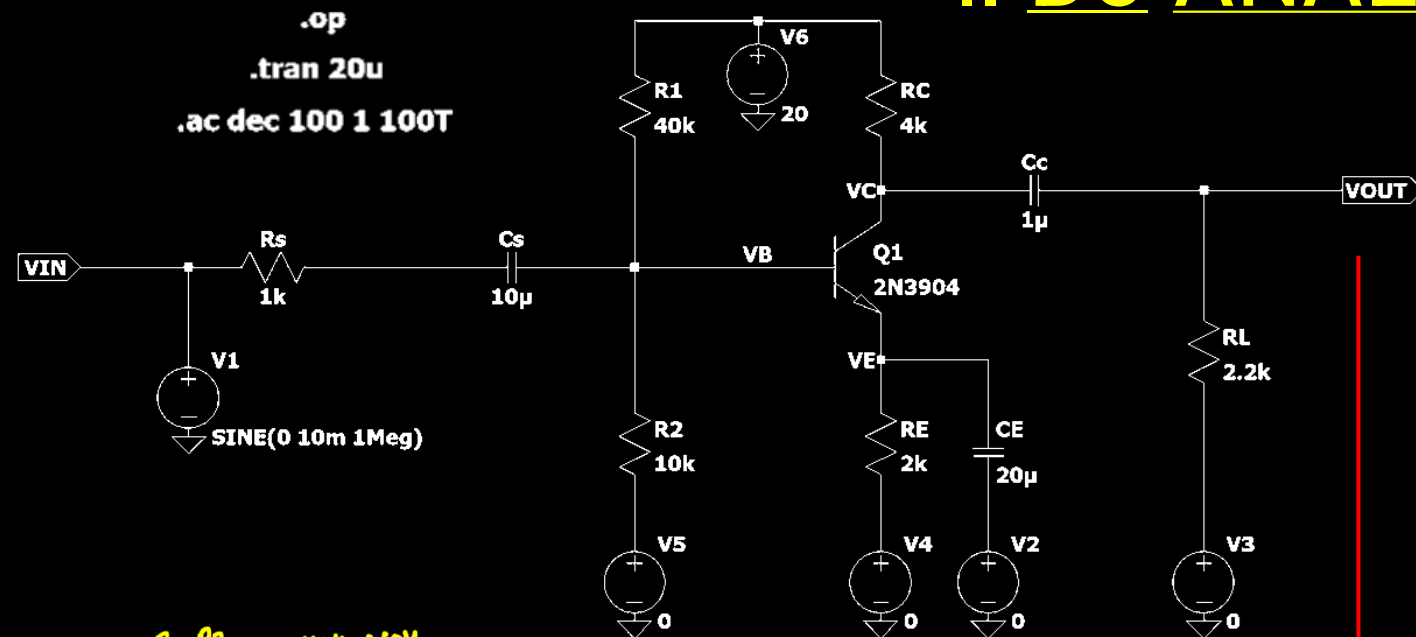


TRANSFORMER COUPLED AMPLIFIER



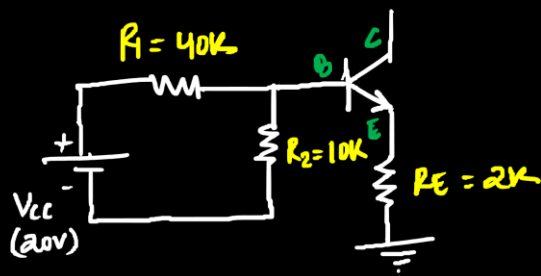
DIRECT COUPLED AMPLIFIER

I. DC ANALYSIS



$$R_{th} = \frac{R_1 \cdot R_2}{R_1 + R_2} = \frac{40k \times 10k}{40k + 10k}$$

$$R_{th} = 8k \checkmark$$

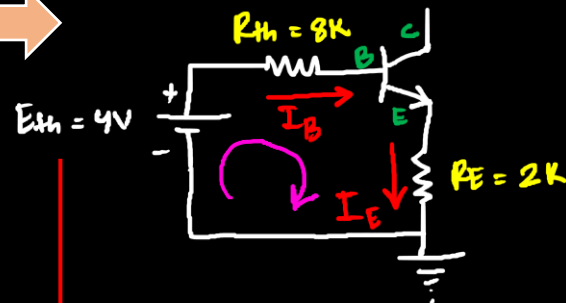


$$V_{th} = V_B = \frac{V_{cc} \cdot R_2}{R_1 + R_2} = \frac{20V \times 10k}{10k + 40k}$$

$$V_{th} = V_B = 4V.$$

> 0.7V
to TURN ON the BJT.

Putting KVL ; Wegel



$$V_{th} - I_B \cdot R_{th} - V_{BE} - I_E \cdot R_E = 0$$

$$I_B = \frac{V_{th} - V_{BE}}{R_{th} + (\beta + 1) \cdot R_E} = \frac{20V - 0.7V}{8k + (90 + 1) \cdot 2k} = \frac{19.3V}{8k + 182k} = 0.1015 \text{ mA}$$

$$\Rightarrow I_B = 5.78 \mu A \checkmark$$

kle know;

$$I_C = \beta \cdot I_B$$

$$= 90 \times 5.78 \mu A$$

$$= 520.2 \mu A$$

$$I_C = 0.5202 \text{ mA}$$

$$I_E = I_B + I_C$$

$$= 5.78 \mu A + 0.5202 \text{ mA}$$

$$= 0.52598 \text{ mA}$$

$$I_E = 0.525 \text{ mA}$$

$$\text{Again ; } V_{CE} = V_C - V_E = V_{cc} - I_C \cdot R_C - I_E \cdot R_E$$

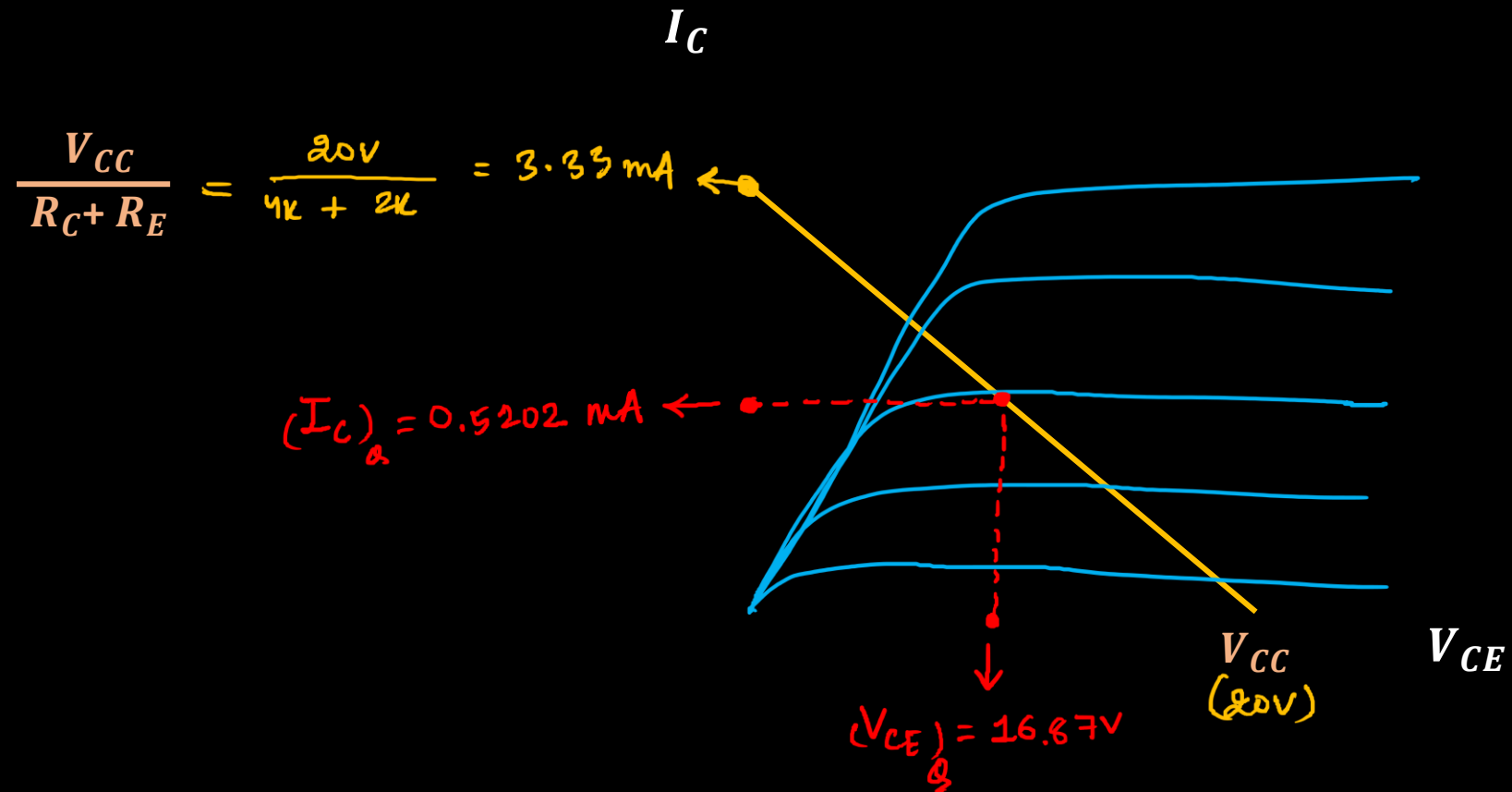
$$= 20V - (0.5202 \text{ mA} \times 4k) - (0.525 \text{ mA} \times 2k)$$

$$V_{CE} = 20V - 2.08V - 1.05V$$

$$V_{CE} = 16.87V \checkmark$$

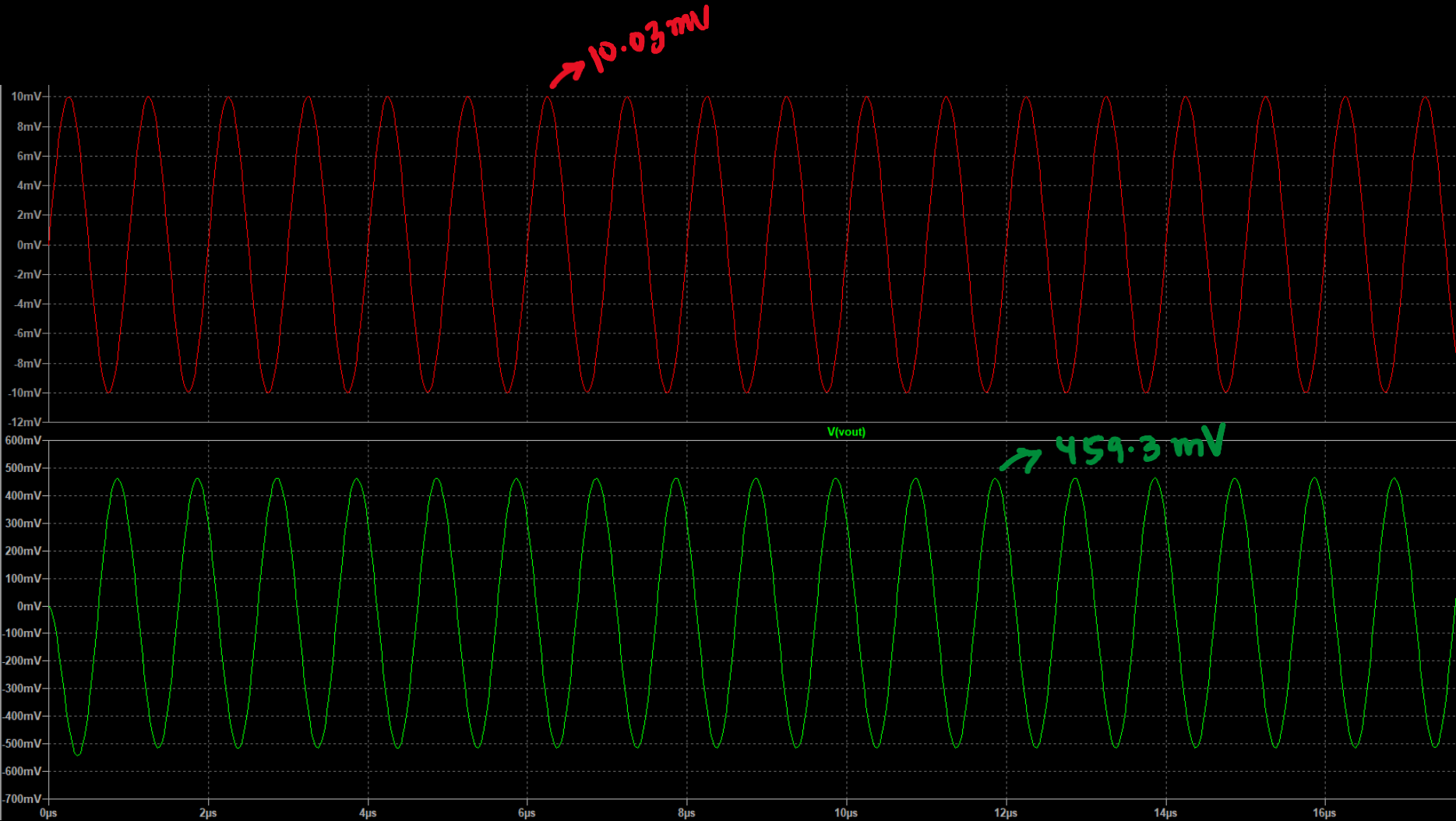
--- Operating Point ---

V(vb) :	3.95985	voltage
V(n001) :	20	voltage
V(n003) :	0	voltage
V(n002) :	3.95985e-14	voltage
V(n005) :	0	voltage
V(vc) :	13.4323	voltage
V(ve) :	3.29389	voltage
V(n004) :	0	voltage
V(vout) :	2.95511e-14	voltage
V(n006) :	0	voltage
V(vin) :	0	voltage
Ic(Q1) :	0.00164193	device_current
Ib(Q1) :	5.01825e-06	device_current
Ie(Q1) :	-0.00164694	device_current
Is(Q1) :	0	device_current
I(RE) :	0.00164694	device_current
I(RL) :	1.34323e-17	device_current
I(V5) :	0.000395985	device_current
I(V1) :	3.95985e-17	device_current
I(R1) :	0.000401004	device_current
I(Cs) :	3.95985e-17	device_current
I(V3) :	1.34323e-17	device_current
I(V6) :	-0.00204293	device_current
I(R2) :	0.000395985	device_current
I(Cc) :	-1.34323e-17	device_current
I(RC) :	0.00164193	device_current
I(CE) :	6.58778e-17	device_current
I(V2) :	6.58778e-17	device_current
I(V4) :	0.00164694	device_current
I(Rs) :	3.95985e-17	device_current



D.C Operating Point

II. TRANSIENT ANALYSIS



$$A_v = \text{voltage gain} = \frac{(v_{out})_{pp}}{(v_{in})_{pp}} =$$

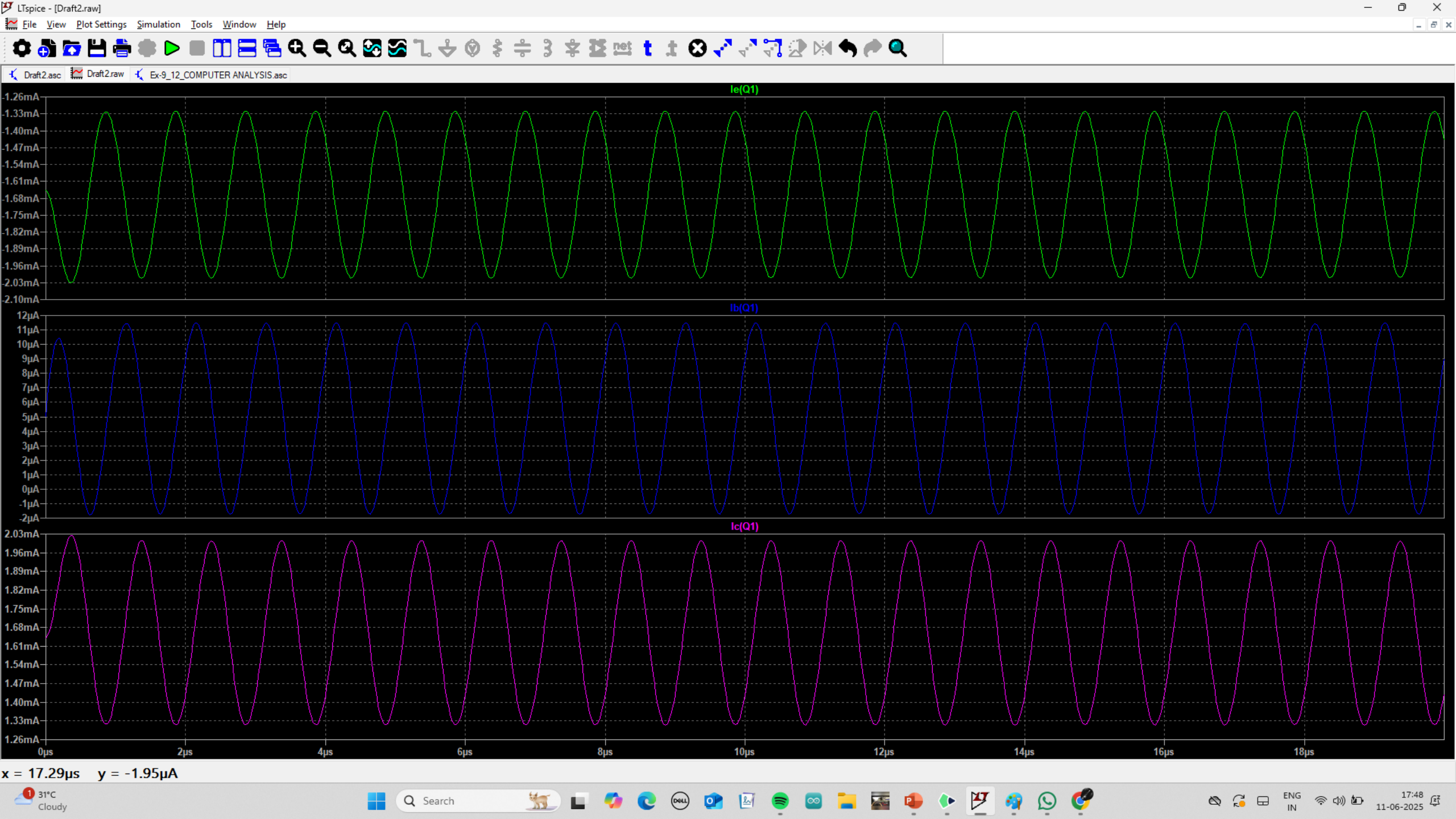
$$A_v = \frac{462.25 \text{ mV} + 514.53 \text{ mV}}{9.992 \text{ mV} + 9.994 \text{ mV}}$$

$$A_v = 48.8732$$

$$A_v = \frac{v_{out}}{v_{in}}$$

$$v_{out} = A_v \times v_{in}^{**}$$

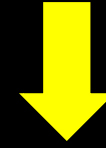
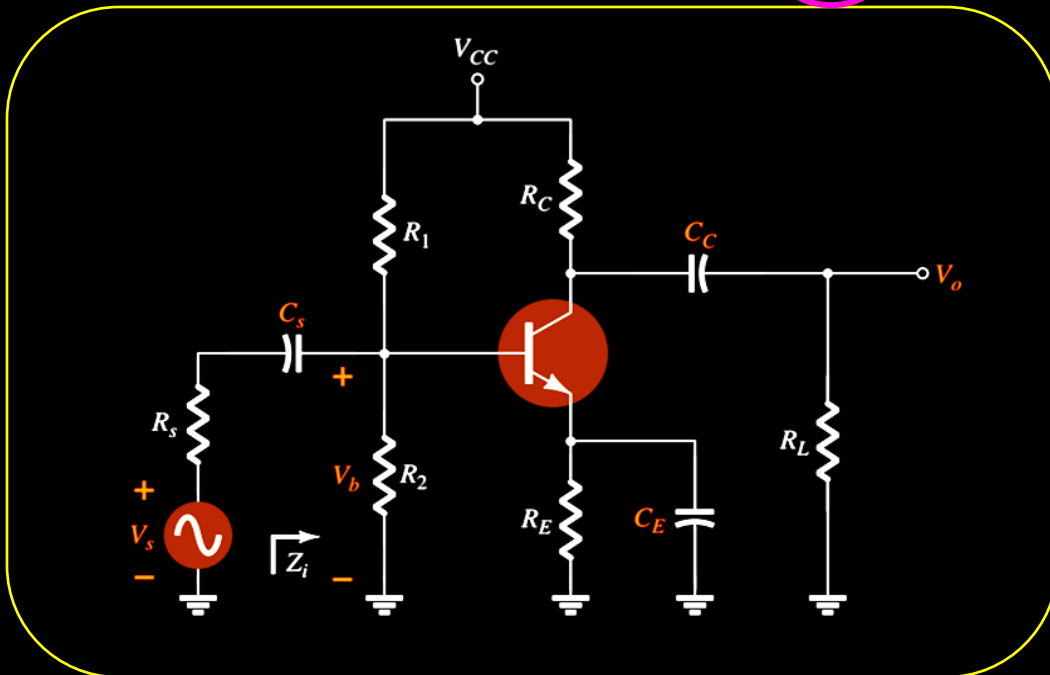
AMPLIFIED & INVERTED OUTPUT



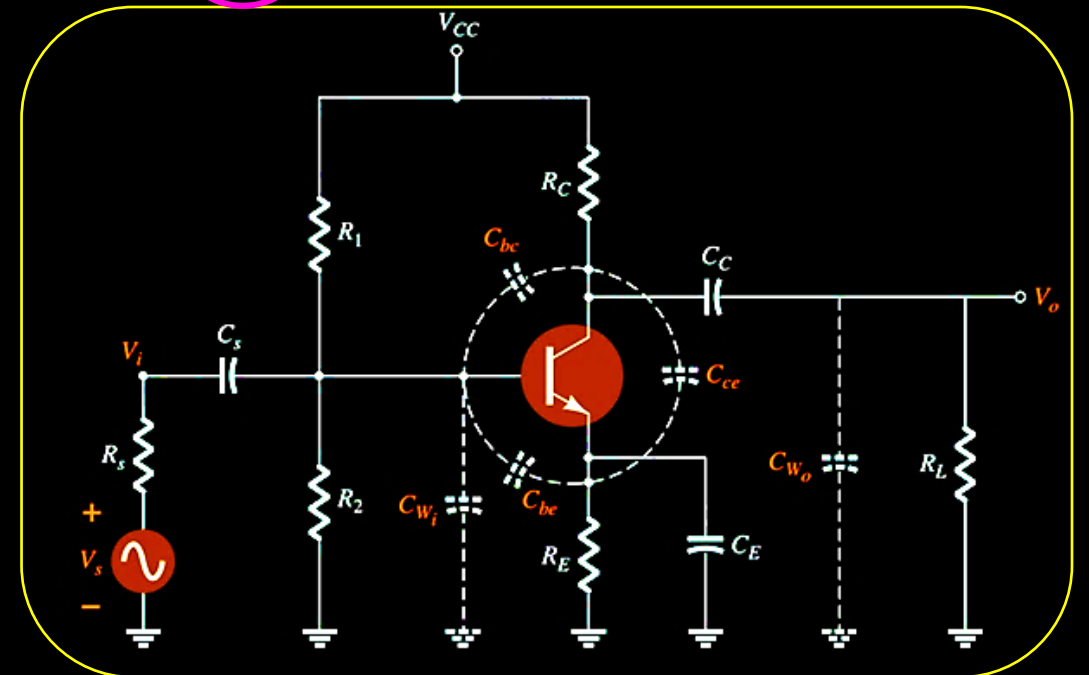
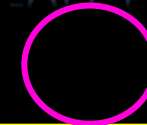
III. AC ANALYSIS

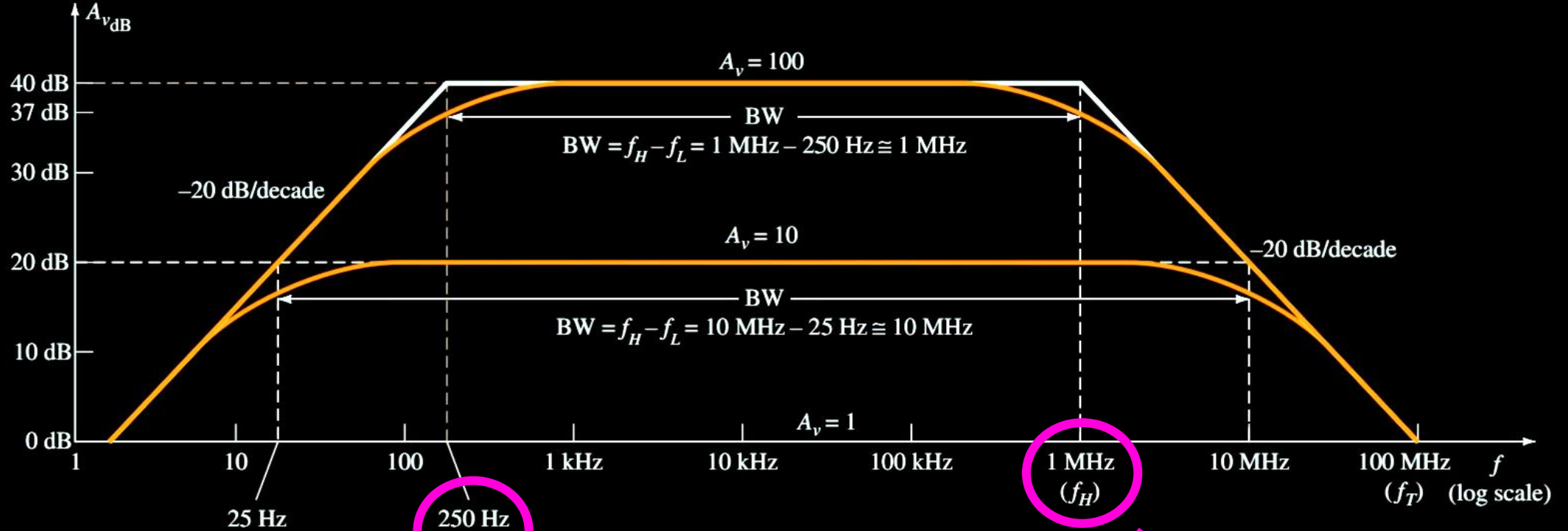


LOW FREQUENCY
RESPONSE
For
Finding the
LOW CUT-OFF FREQUENCY



HIGH FREQUENCY
RESPONSE
For
Finding the
HIGH CUT-OFF FREQUENCY





1. Due to Source Capacitor

$$f_{L_s} = \frac{1}{2\pi(R_s + R_i)C_s}$$

2. Due to By-pass Emitter Capacitor

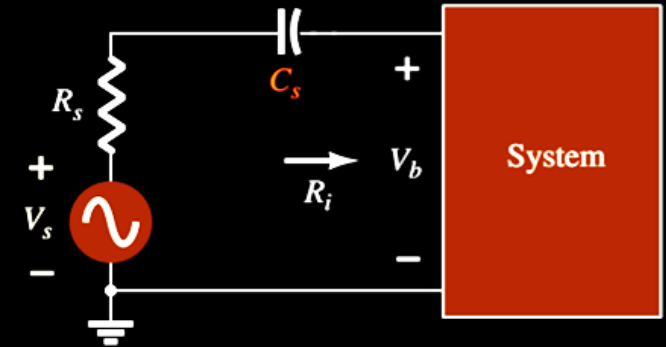
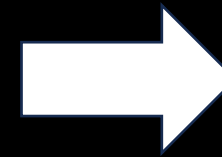
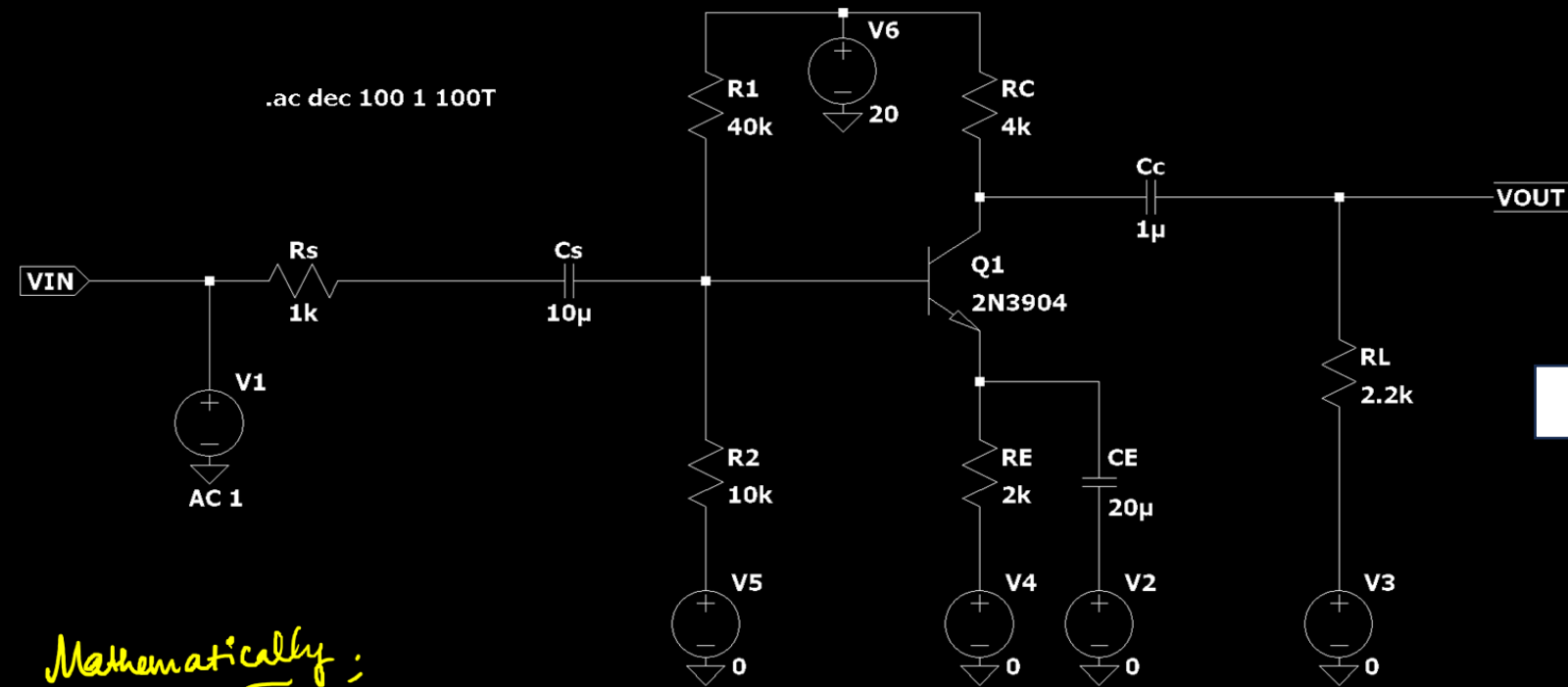
$$f_{L_E} = \frac{1}{2\pi R_e C_E}$$

3. Due to Output Coupling Capacitor

$$f_{L_C} = \frac{1}{2\pi(R_o + R_L)C_C}$$

$$f_{H_i} = \frac{1}{2\pi R_{Th_i} C_i}$$

$$f_{H_o} = \frac{1}{2\pi R_{Th_o} C_o}$$



Mathematically ;

$$A_v = \text{voltage gain} = \frac{V_o}{V_i} = -\frac{R_c \parallel R_L}{r_e} = \frac{R_c \times R_L}{R_c + R_L} \times \frac{1}{r_e} = R_c \parallel R_L \times \frac{1}{V_T / I_E} = R_c \parallel R_L \times \frac{I_E}{V_T} = \frac{1}{V_T} \left[R_c \parallel R_L \times \underbrace{I_E}_{= I_B + I_C} \right]$$

$$A_v \propto R_c \cdot R_L \propto I_E \propto I_S$$

$$= I_B + I_C \cdot e^{\frac{V_{BE}}{V_T}}$$

Since; $V_T = 0.0259 \text{ V} \approx 26 \text{ mV}$ is constant ✓

$$I_C \propto I_S \propto e^{V_{BE}/V_T}$$

That Means; if $e^{V_{BE}} \uparrow \rightarrow I_S \uparrow \rightarrow I_C \uparrow \rightarrow I_E \uparrow \rightarrow r_e \downarrow \rightarrow A_v \uparrow \dots$

$$e^1 = 2.7182 ; e^2 = 7.389$$

So; According to the Model parameters of NXP Semiconductor's → BJT Model Name: "2N3904"

$$\beta = 90$$

$$I_C = ?$$

$$I_C = I_S \cdot e^{V_{BE}/V_T}$$

$$V_{BE} = V_T \cdot \ln\left(\frac{I_C}{I_S}\right) = 0.0259V \cdot \ln\left(\frac{0.5202mA}{10fA}\right) = \boxed{0.692V}$$

$$V_{BE} = 0.692V \text{ or } 692mV$$

We Calculated
taking $\div V_{BE} = 0.7V$.

$$\frac{1}{0.0259V} \cdot \left[\frac{(R_C \times R_L)}{R_C + R_L} \cdot (I_E) \right]$$

$$= \frac{1}{0.0259V} \cdot \left[\frac{4k \times 2.2k}{4k + 2.2k} \cdot (1.654mA) \right]$$

=

But According to our DC Biased operating point:-

$$V_{th} = V_{cc} \frac{R_2}{R_1 + R_2} = \frac{20V \cdot 10k}{40k + 10k} = \frac{200}{50} = \boxed{4V}$$

$$V_{th} = V_B = 4V$$

$[V_{th} \text{ or } V_B > V_{BE}] \rightarrow$ Hence; Transistor is ON !!

$$\left. \begin{array}{l} V_{CE} = 16.87V \\ I_C = 0.5202mA \end{array} \right\} \text{SUFFICIENTLY HIGH } \checkmark$$

Hence [ACTIVE
REGION]
=.

Now; Let's put the values and find out $\bar{A}_v'$ [voltage gain]

$$A_v = \frac{V_o}{V_i} = -\frac{R_C \parallel R_L}{r_e} = \frac{1}{V_T} \cdot [(R_C \parallel R_L) \cdot I_E]$$